

Some Basics of Molecular Biology

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ETH

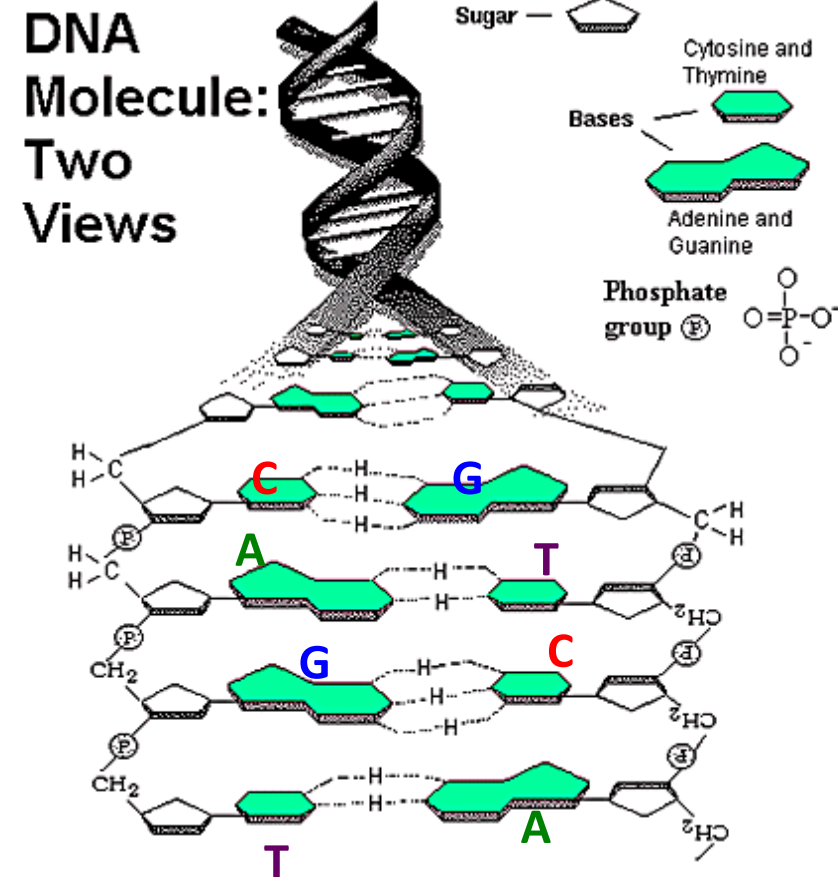
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Messages

- DNA / genome size /evolution
- genes – concept
- gene regulation
- RNA species; junk
- RNA numbers
- RNA degradation

What is DNA?

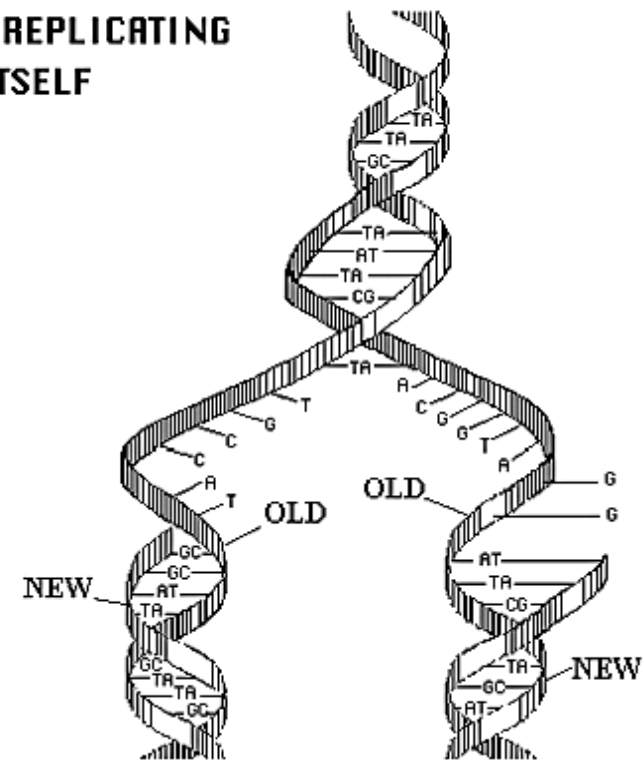
- A long backbone of sugars with nucleotides attached
 - **Adenine (A)**
 - **Guanine (G)**
 - **Cytosine (C)**
 - **Thymine (T)**
- It can form a self-complementary **double helix**
- In living organisms, the DNA is the carrier of the hereditary information, it is the **source code** of life



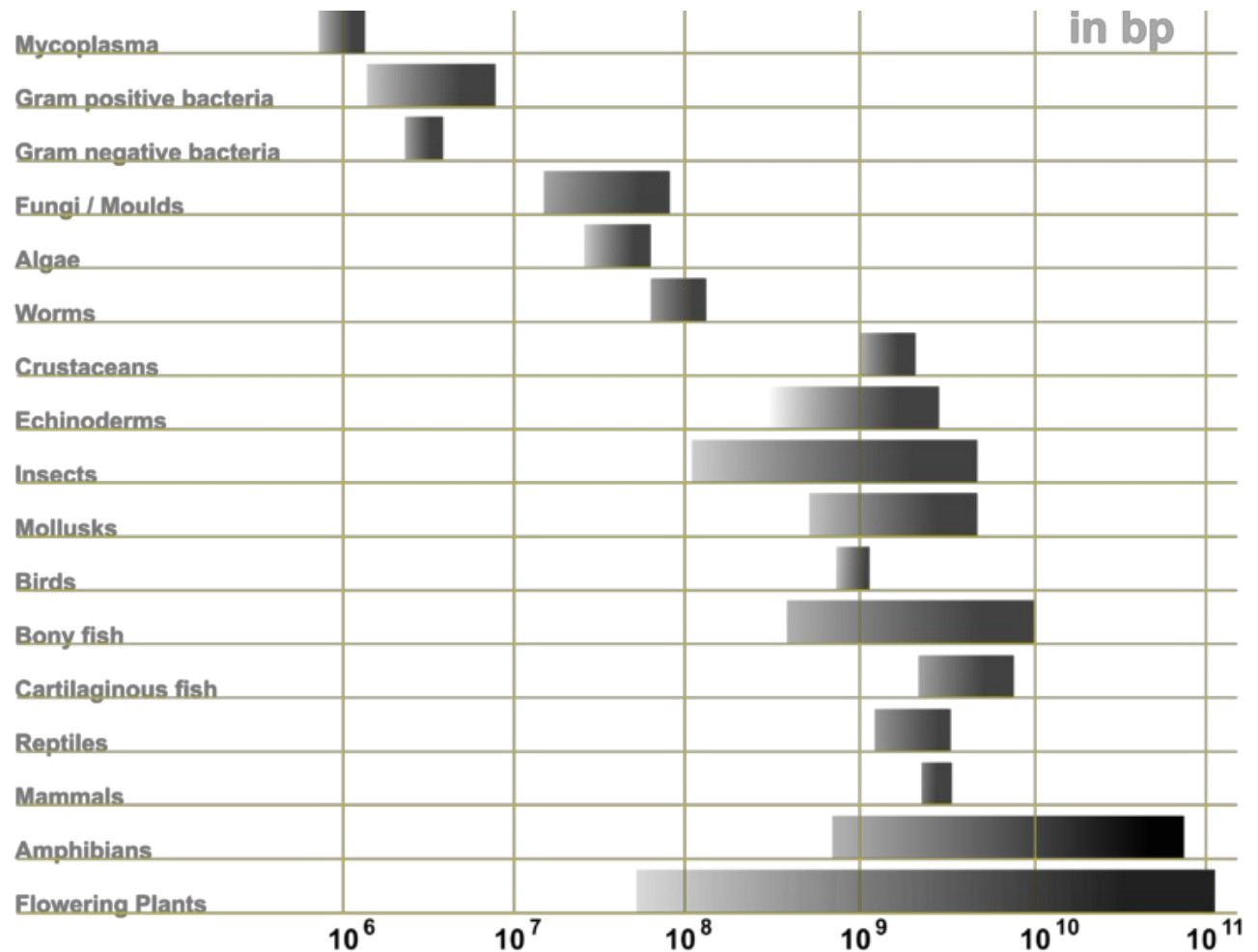
DNA replication

- The helix becomes unzipped and each strand acts as a template for a new complementary strand of DNA

DNA REPLICATING ITSELF



Genome Sizes



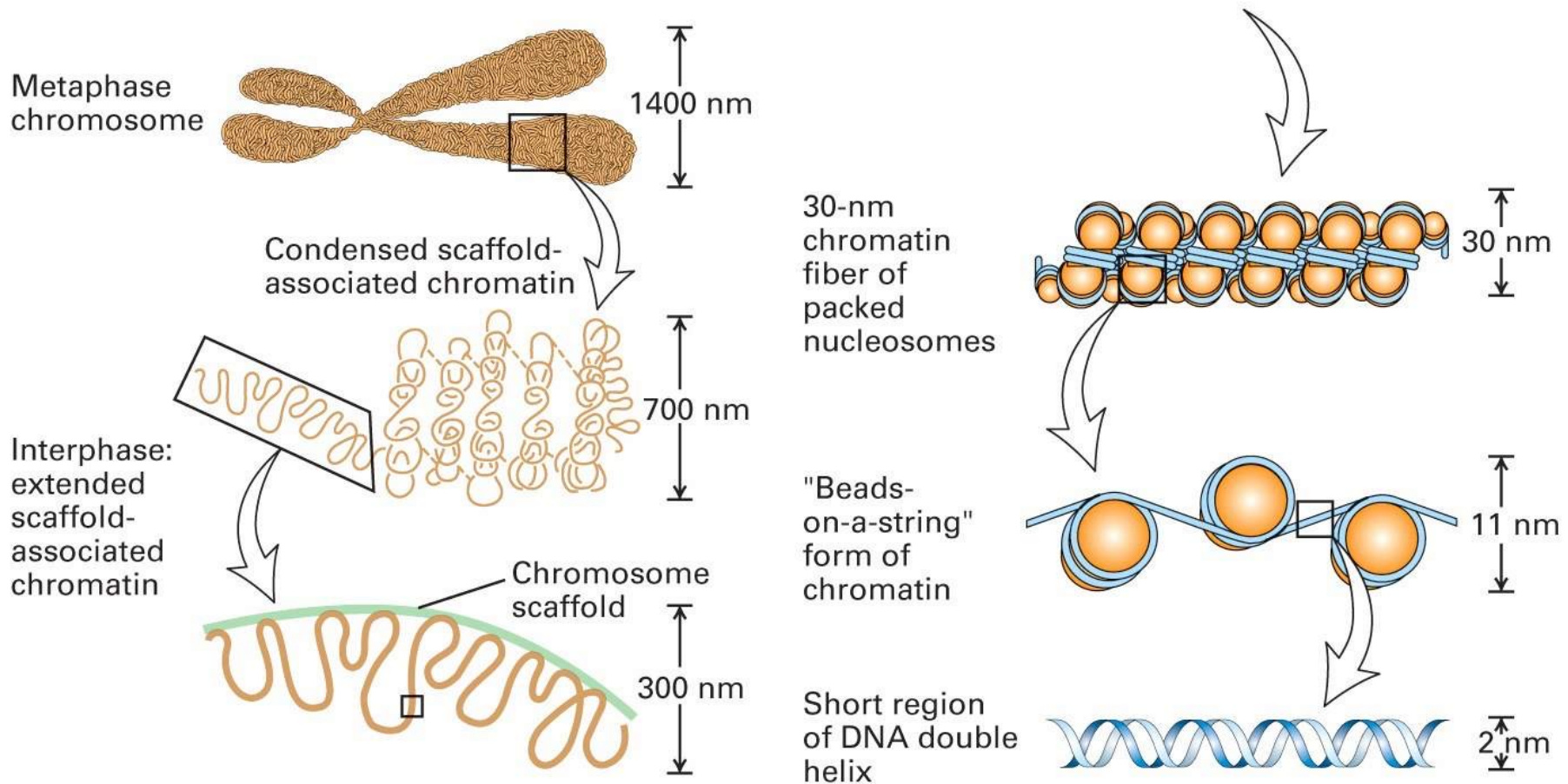
The size of the human genome is 3.1 billion base pairs (T2T-CHM13, 2022: 3.055 Gb, gap-free)

The length of the entire DNA in a cell is approx. 2m. (diploid)

Genome Size and Complexity: the C-value Paradox

- larger genomes and complex organisms have emerged by **evolution**
- “**Nothing in Biology Makes Sense Except in the Light of Evolution**” — Theodosius Dobzhansky, 1973
- genome sizes vary enormously — in some cases even between closely related species
- genome size is **not** proportional to organismal complexity → the **C-value paradox**
 - what varies is mostly non-coding content: repeats, transposable elements, intron length
- the number of protein-coding genes is remarkably constant **across animals** (~20 000)
 - C. elegans: 959 somatic cells and ~20 000 genes — about as many as a human
 - but not across all life: yeast ~6 000, Arabidopsis ~27 000
- one gene ≠ one protein: alternative splicing, RNA editing, post-translational modification

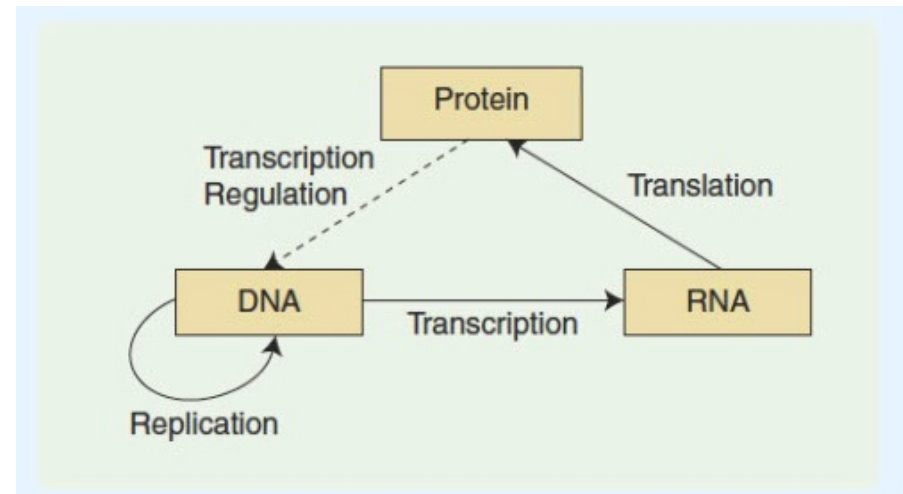
DNA Superstructure



Lodish et al. *Molecular Cell Biology* (5th ed.). W.H. Freeman & Co., 2003.

Genes

- A gene is a region of DNA (a locus) that controls a hereditary characteristic
- protein coding genes are transcribed into messenger RNA which is then translated into protein.
- In humans the exons of protein-coding genes cover only ~2% of the genome (coding sequence alone ~1.5%) — but the gene loci themselves, introns included, cover ~25–45% depending on the annotation

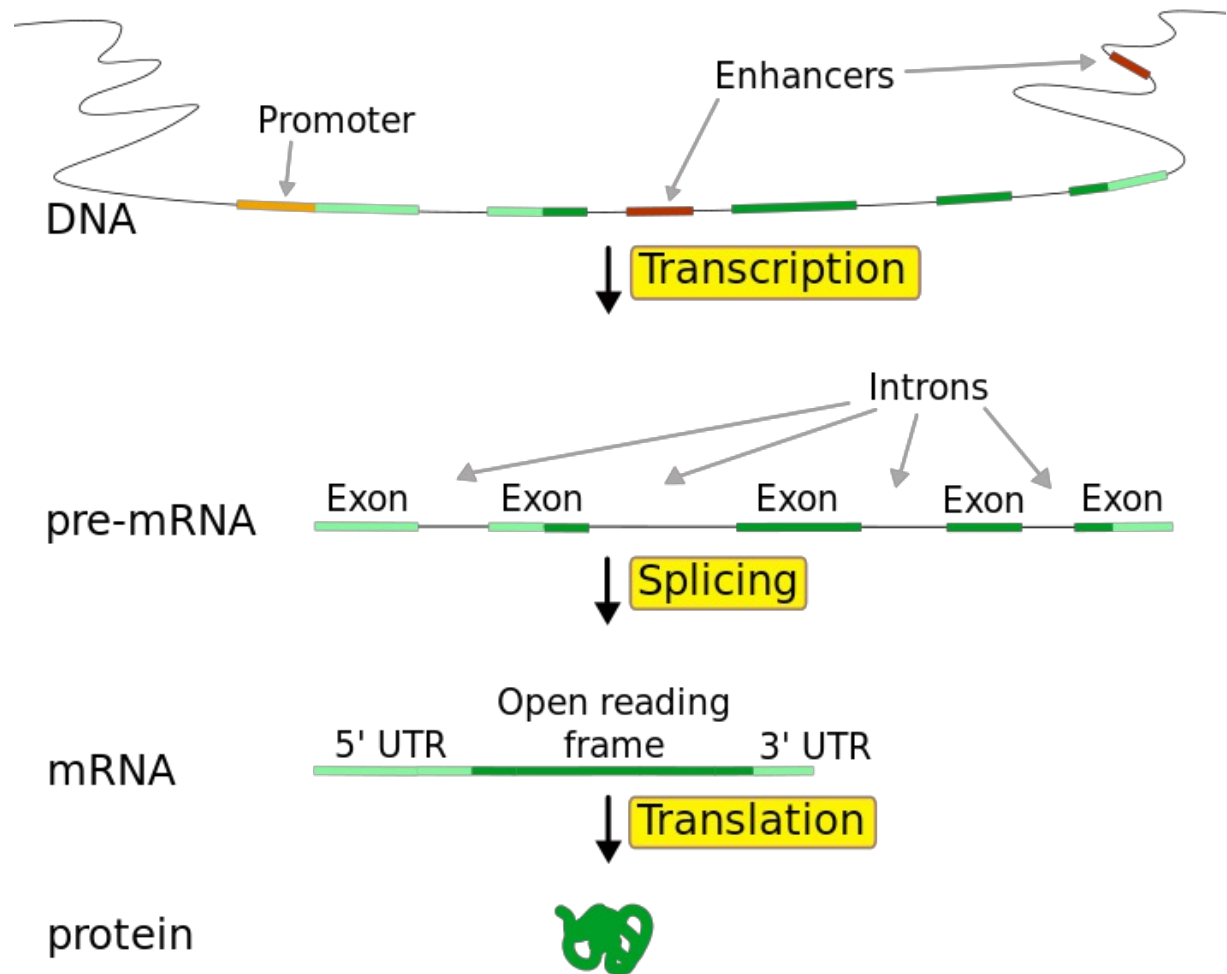


Transcription

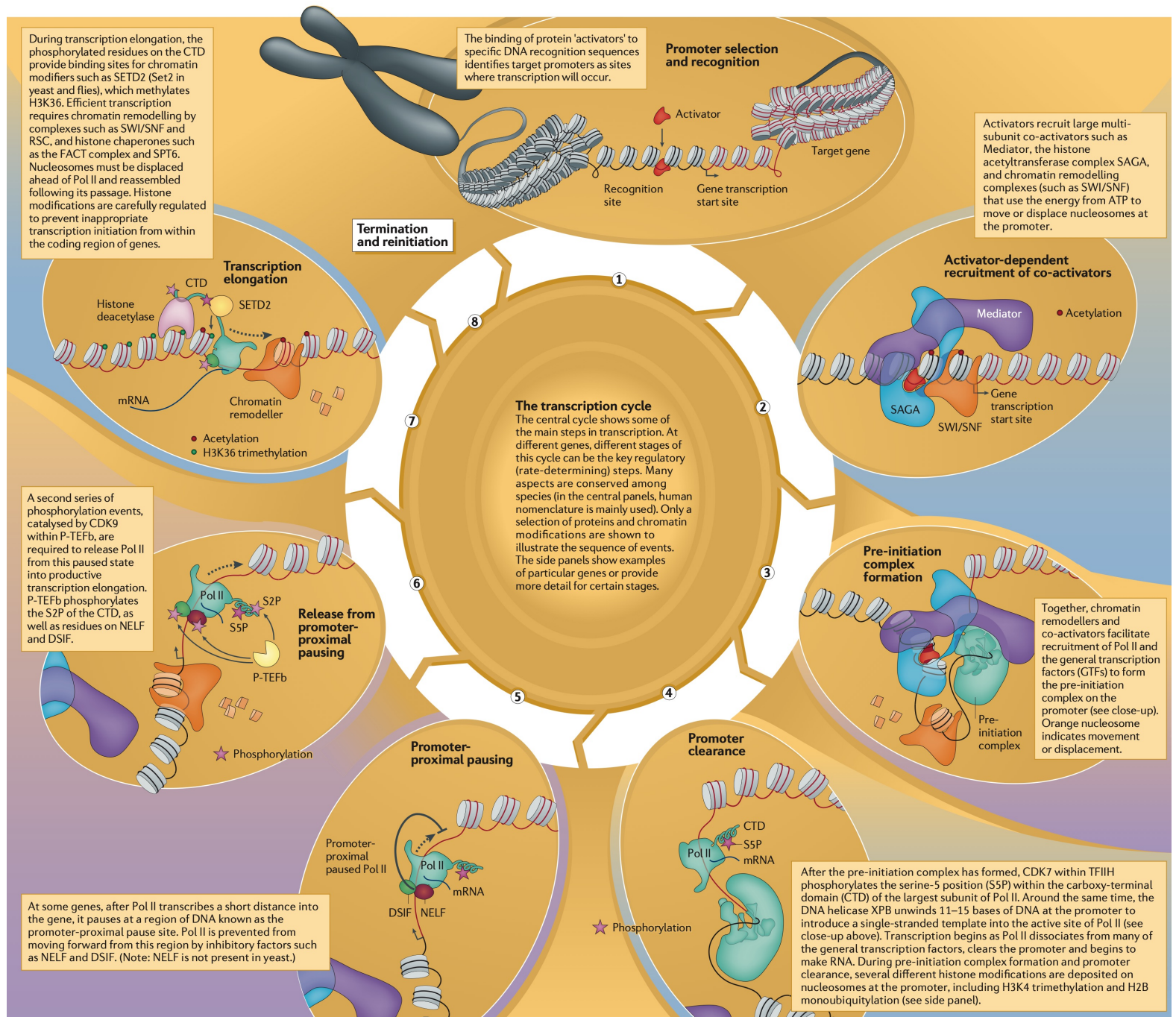
The transcription process generates a messenger RNA molecule from a gene region. RNA is like DNA but

- the sugar-phosphate is different: ribose instead of deoxyribose
- In all places where the DNA has a T the the RNA has a U (uracil)

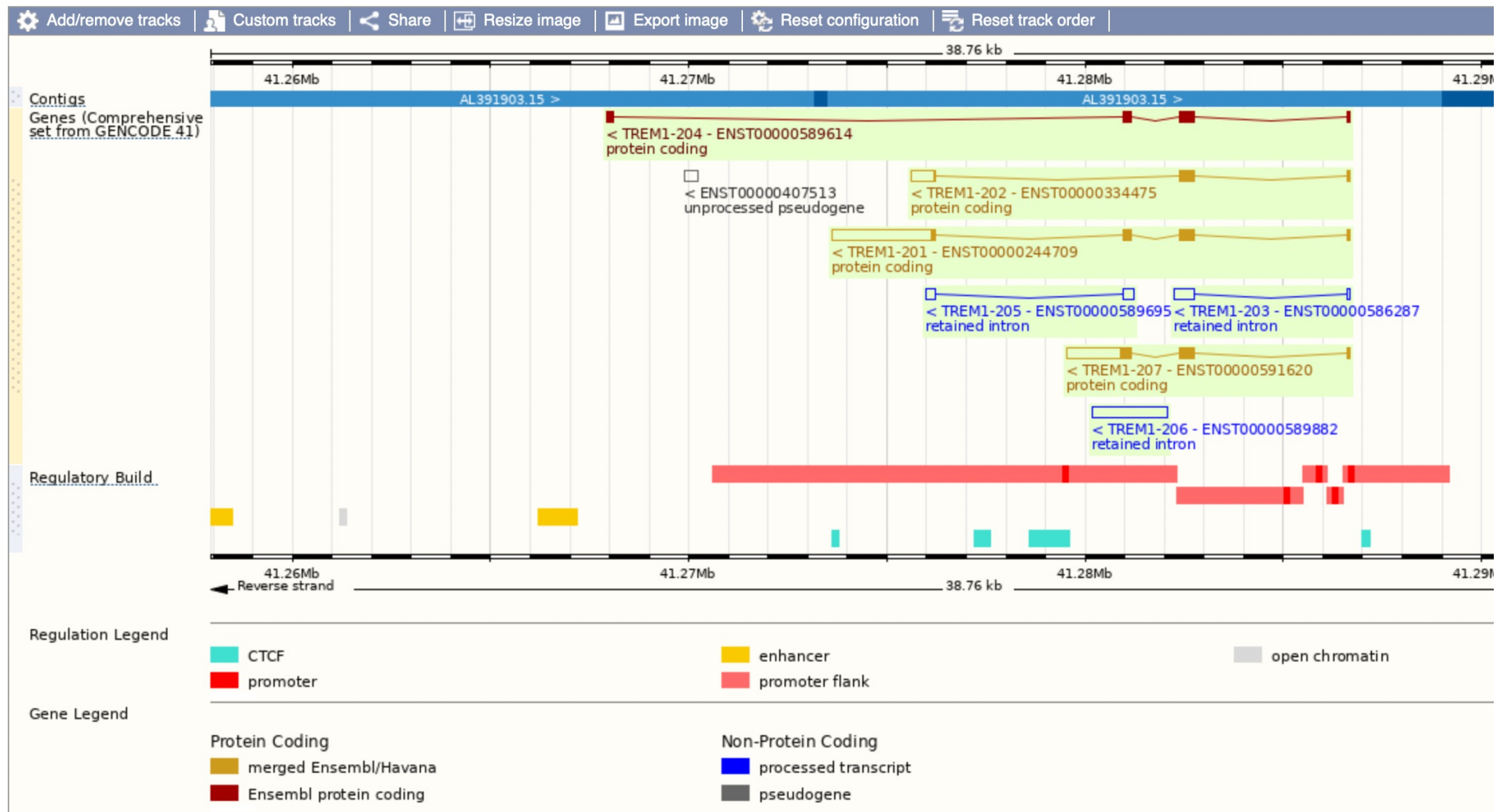
In higher organisms the protein coding sequences (exons) are interspersed by non-coding sequences (introns) which are spliced out.



Transcription Cycle



Gene Structure

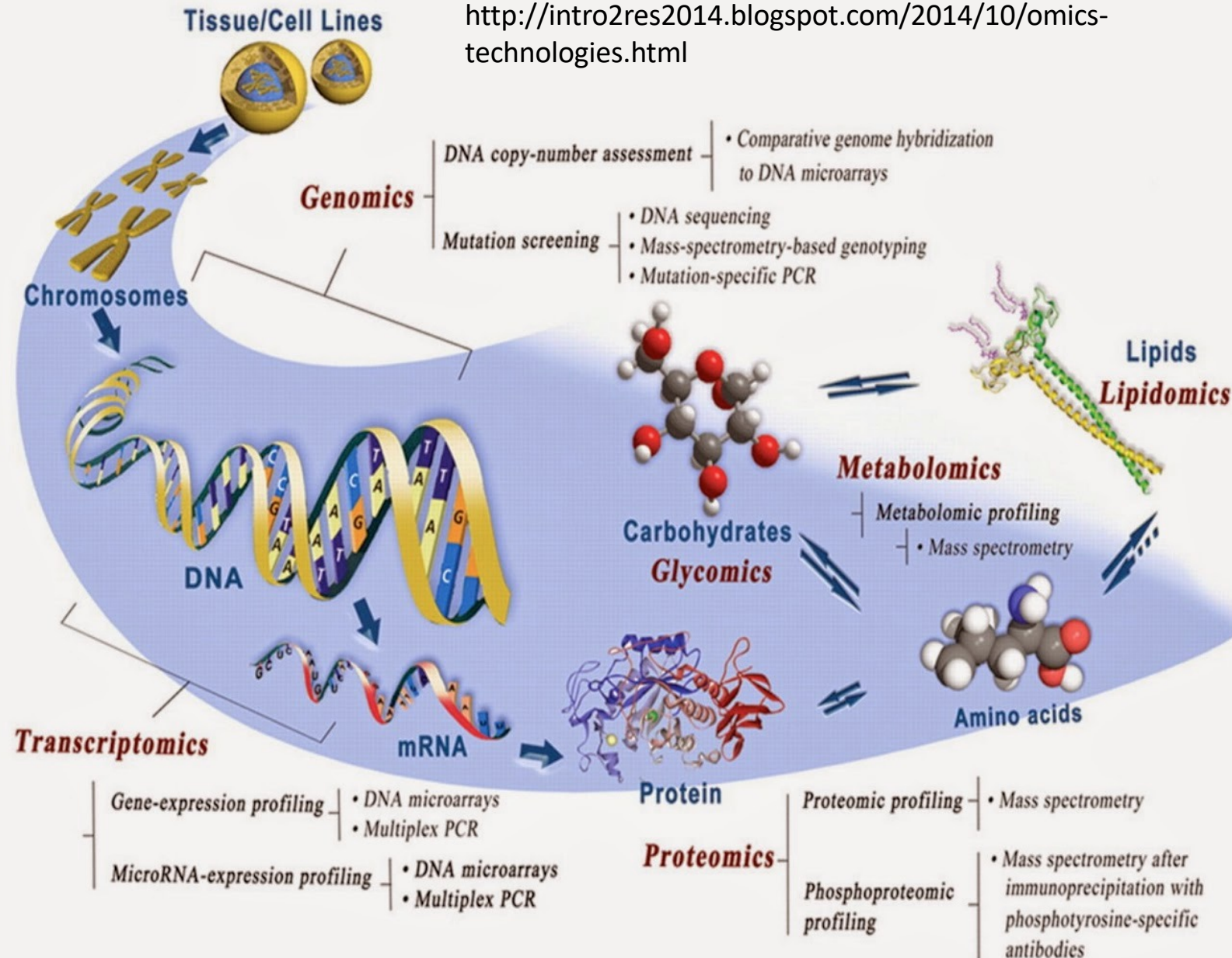


- introns are much longer than exons
- humans: typical gene length(exonic): 1000 – 2000 bases
- Genome Browsers: UCSC, Ensembl, IGV

Gene Regulation: Where the Signal Comes From

- a gene's transcription rate is set by proteins binding DNA near it — and far from it
 - **promoter** — where RNA polymerase assembles, at the transcription start site (TSS)
 - **enhancers / silencers** — regulatory sequence that can sit 10 kb to 1 Mb away, inside introns or beyond neighbouring genes
 - **transcription factors** — proteins that read those sequences and recruit or block the polymerase
 - **chromatin state** — DNA that is wound up is not available to be read at all
- every cell carries the same genome; different regulators are bound → different transcriptomes
 - this is why a cell type is defined by its expression profile, not by its DNA

<http://intro2res2014.blogspot.com/2014/10/omics-technologies.html>



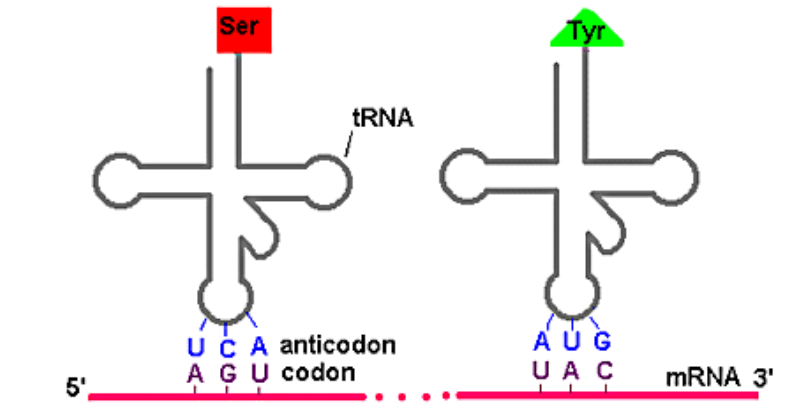
DNA/RNA vs proteins & metabolites

- DNA/RNA are linear chains of nuclear acids
 - DNA/RNA can have secondary structure but can linearized
 - The chains can be sequenced, (can be read sequentially)
 - Identification and quantification can be performed by sequencing
 - Note: Quantification can additionally be performed by specific binding to probes
- Proteins and metabolites often have 3D structure
 - can be **identified** by their specific mass and the masses of their fragments
 - can be **quantified** by binding assays if they have specific binding properties that are known

Translation: The Genetic Code

The translation process generates a protein based on the information in the messenger RNA

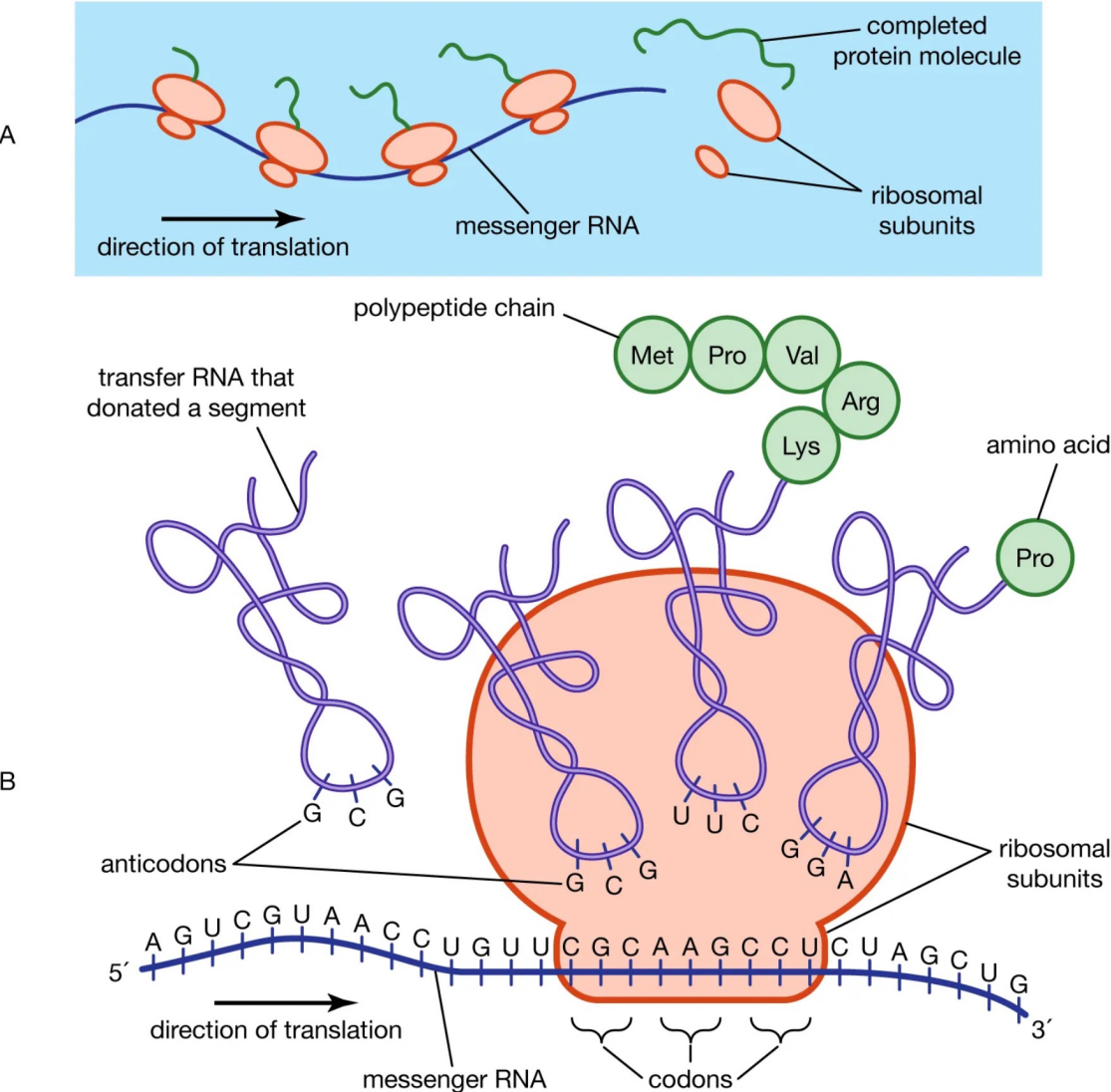
- A protein is a linear polymer of amino acids linked together by peptide bonds.
- Proteins are the main functional chemicals in the cell, carrying out many functions, for example catalysis of the reactions involved in metabolism.
- Proteins have a complex spatial structure



		2nd base in codon					
		U	C	A	G		
1st base in codon	U	Phe Phe Leu Leu	Ser Ser Ser Ser	Tyr Tyr STOP STOP	Cys Cys STOP Trp	U C A G	3rd base in codon
	C	Leu Leu Leu Leu	Pro Pro Pro Pro	His His Gln Gln	Arg Arg Arg Arg	U C A G	
	A	Ile Ile Ile Met	Thr Thr Thr Thr	Asn Asn Lys Lys	Ser Ser Arg Arg	U C A G	
	G	Val Val Val Val	Ala Ala Ala Ala	Asp Asp Glu Glu	Gly Gly Gly Gly	U C A G	

The Genetic Code

Translation



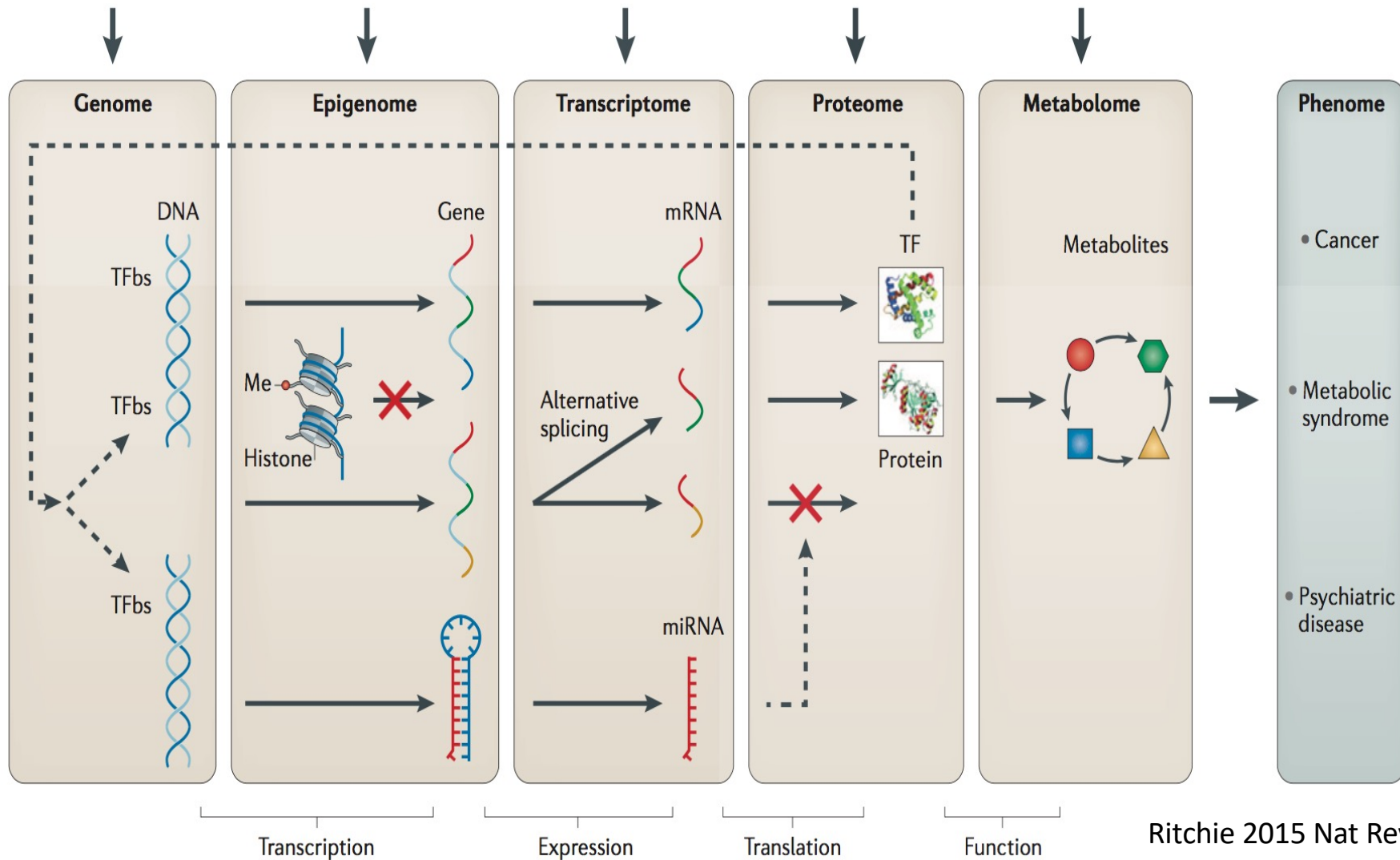
- SNP
- CNV
- LOH
- Genomic rearrangement
- Rare variant

- DNA methylation
- Histone modification
- Chromatin accessibility
- TF binding
- miRNA

- Gene expression
- Alternative splicing
- Long non-coding RNA
- Small RNA

- Protein expression
- Post-translational modification
- Cytokine array

- Metabolite profiling in serum, plasma, urine, CSF, etc.



NGS Reads

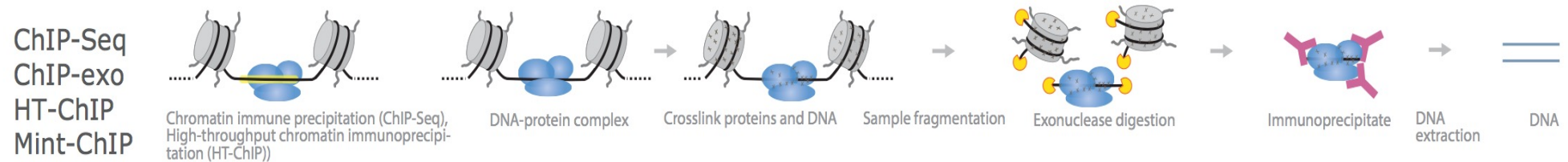


- Extract nucleotide material from specimens
- Select molecules
 - this is the crucial step that determines what information we can learn later from the data
 - Examples: mRNA; mRNA that is currently being translated; DNA positions (promoter regions) where a protein (transcription factor) is bound; cell-free RNA; ...
- Add adapters/primers and make the molecules compatible with the sequencer
- Sequencing: Identify the nucleotide sequences

NGS Protocols

- Example: Preparation of DNA for a ChIP-seq experiment

DNA-Protein Interactions



- The preparation determines how sequenced reads have to be interpreted

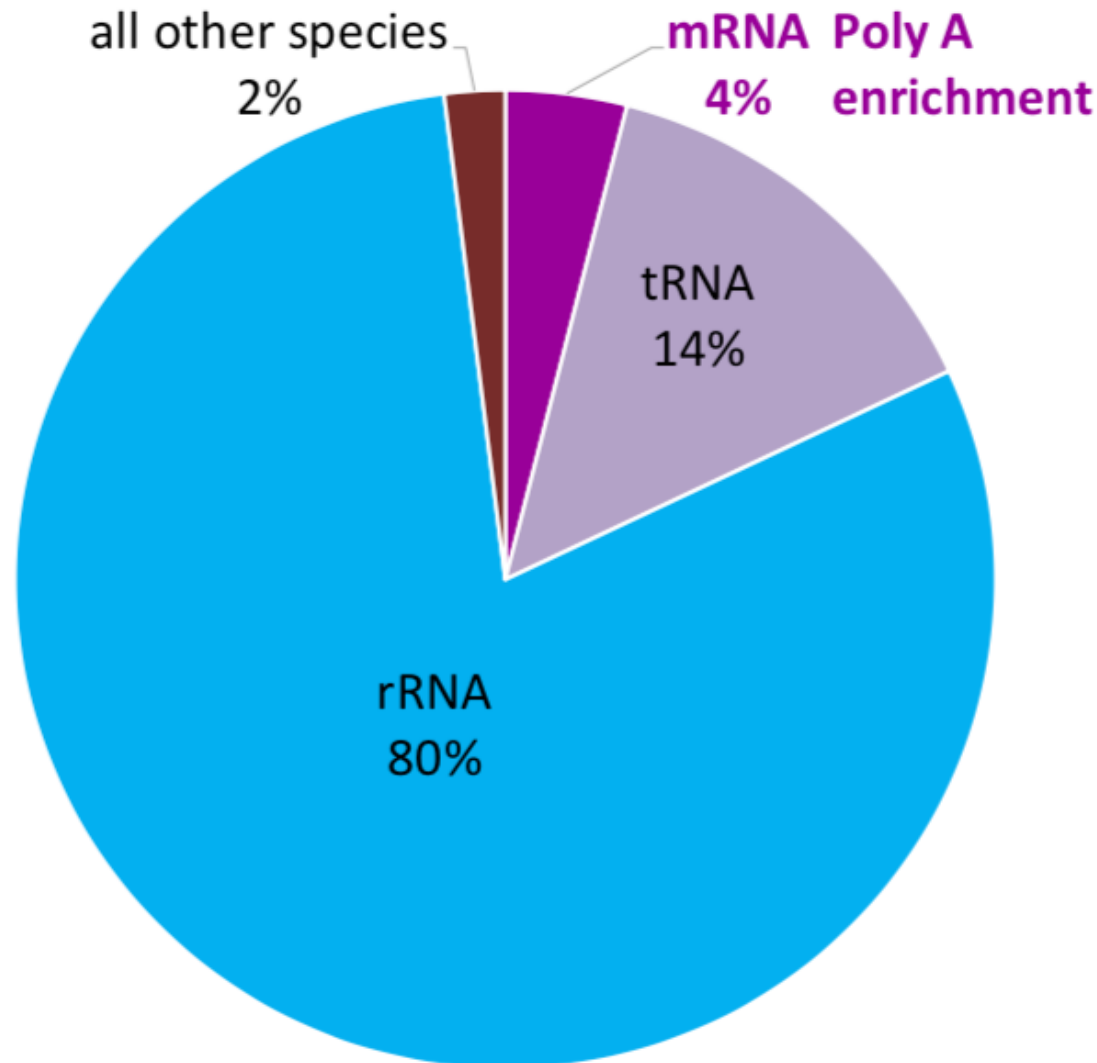
- DNA variants
- DNA modifications (Epigenetics)
- DNA–protein interactions
- Protein–protein interactions
- RNA abundance (gene expression)
- RNA transcription
- RNA–protein interactions
- RNA modifications
- RNA structure

- Big Challenge: targeted molecules may degrade during extraction

RNA-seq in numbers

- Typical numbers for human cells
- In biology everything is possible!
Be prepared to encounter cells that have 10x less or 10x more.
- 250'000 – 500'000 mRNA molecules per cell
- ~14% of total RNA is in nucleus
- RNA:DNA in nucleus 1:2
- typical mRNA molecule length ~1900nt

Prevalence of RNA species



- transfer RNA (tRNA)
length: 76-90 nt
- ribosomal RNA (rRNA),
different subunits
 - 5S: 121 nt
 - 5.8S: 156 nt
 - 18S: 1869 nt
 - 28S: 5070 nt
- plot shows shares of masses

mRNA abundance

Abundance class	Copies/cell	Number of different messages/cell	Abundance of each message
Low	5–15	11,000	<0.004%
Intermediate	200–400	500	<0.1%
High	12,000	<10	3%

- A highly expressed gene typically does not contribute more than 3% of all transcripts; in extreme cases it may be more than 20%
- Large dynamic range

RNA-seq experiment of bulk tissue

- ~ 1 Mio cells as input, this gives > 10ug RNA material
- used as input for RNA-seq protocol: 100 – 1000 ng of total RNA
- > =100 billion molecules as starting material; only a small fraction of the molecules is mRNA
- Steps:
 - Poly-A based enrichment oligo(dT) with beads
 - fragmentation
 - PCR Amplification (e.g. ~7 – 20 cycles)
 - 10 cycles: every molecule has ~1000 duplicates
 - sequencing (sampled readout of ~20 Mio processed molecules)
- Note: special protocols attach a random sequence before amplification
 - Serves as unique molecule identifier (UMI)
 - Recommended for low input cases, e.g. starting amount < 50ng

PCR amplification

- PCR amplification is assumed to amplify any fragment equally well
- BUT PCR can introduce bias
 - Amplification depends on GC content of fragment (extremely high/low GC content is underrepresented)
 - Shorter fragments are amplified more efficiently
 - Long fragments may be amplified incompletely
 - ...
 - Bias depends on number of cycles
 - Bias needs to be kept consistent

Summary

- Sequencing polls DNA/RNA molecules
→ the library preparation determines what is selected and what information can be extracted
- Sequencing is a massive sampling process
→ the mechanics of the sampling determine the analysis steps and the statistical models to be applied
- Analysis need to be aware how the data is obtained